

Features

- High-performance, Low-power Atmel® AVR® 8-bit Microcontroller
- Advanced RISC Architecture
 - 131 Powerful Instructions – Most Single-clock Cycle Execution
 - 32 x 8 General Purpose Working Registers
 - Fully Static Operation
 - Up to 16 MIPS Throughput at 16 MHz
 - On-chip 2-cycle Multiplier
- High Endurance Non-volatile Memory segments
 - 32Kbytes of In-System Self-programmable Flash program memory
 - 1024 Bytes EEPROM
 - 2Kbyte Internal SRAM: static RAM (transistor) [data memory]
 - Write/Erase Cycles: 10,000 Flash/100,000 EEPROM
 - Data retention: 20 years at 85°C/100 years at 25°C⁽¹⁾
 - Optional Boot Code Section with Independent Lock Bits
 - In-System Programming by On-chip Boot Program
 - True Read-While-Write Operation
 - Programming Lock for Software Security
- JTAG (IEEE std. 1149.1 Compliant) Interface → Debugger
 - Boundary-scan Capabilities According to the JTAG Standard
 - Extensive On-chip Debug Support
 - Programming of Flash, EEPROM, Fuses, and Lock Bits through the JTAG Interface
- Peripheral Features
 - Two 8-bit Timer/Counters with Separate Prescalers and Compare Modes
 - One 16-bit Timer/Counter with Separate Prescaler, Compare Mode, and Capture Mode
 - Real Time Counter with Separate Oscillator (WDT)
 - Four PWM Channels → Inside timer PWM Mode
 - 8-channel, 10-bit ADC
 - 8 Single-ended Channels
 - 7 Differential Channels in TQFP Package Only
 - 2 Differential Channels with Programmable Gain at 1x, 10x, or 200x
 - Byte-oriented Two-wire Serial Interface Communication
 - Programmable Serial USART → Communication
 - Master/Slave SPI Serial Interface → Communication
 - Programmable Watchdog Timer with Separate On-chip Oscillator
 - On-chip Analog Comparator
- Special Microcontroller Features
 - Power-on Reset and Programmable Brown-out Detection
 - Internal Calibrated RC Oscillator → generate internal clock. (clock source)
 - External and Internal Interrupt Sources
 - Six Sleep Modes: Idle, ADC Noise Reduction, Power-save, Power-down, Standby and Extended Standby
- I/O and Packages
 - 32 Programmable I/O Lines → Code
 - 40-pin PDIP, 44-lead TQFP, and 44-pad QFN/MLF
- Operating Voltages
 - 2.7V - 5.5V for ATmega32L → low power
 - 4.5V - 5.5V for ATmega32 → normal
- Speed Grades
 - 0 - 8MHz for ATmega32L (0, 1, 2, 3, 4 MHz)
 - 0 - 16MHz for ATmega32 (0, 1, 2, 3, 4 MHz)
- Power Consumption at 1 MHz, 3V, 25°C
 - Active: 1.1mA
 - Idle Mode: 0.35mA
 - Power-down Mode: < 1µA

* → each pin for Atmega provide
 → 5V → Max 40 mA I_{DC} (20:25mA)



8-bit AVR[®]
 Microcontroller
 with 32KBytes Memory
 In-System
 Programmable
 Flash → ROM Code

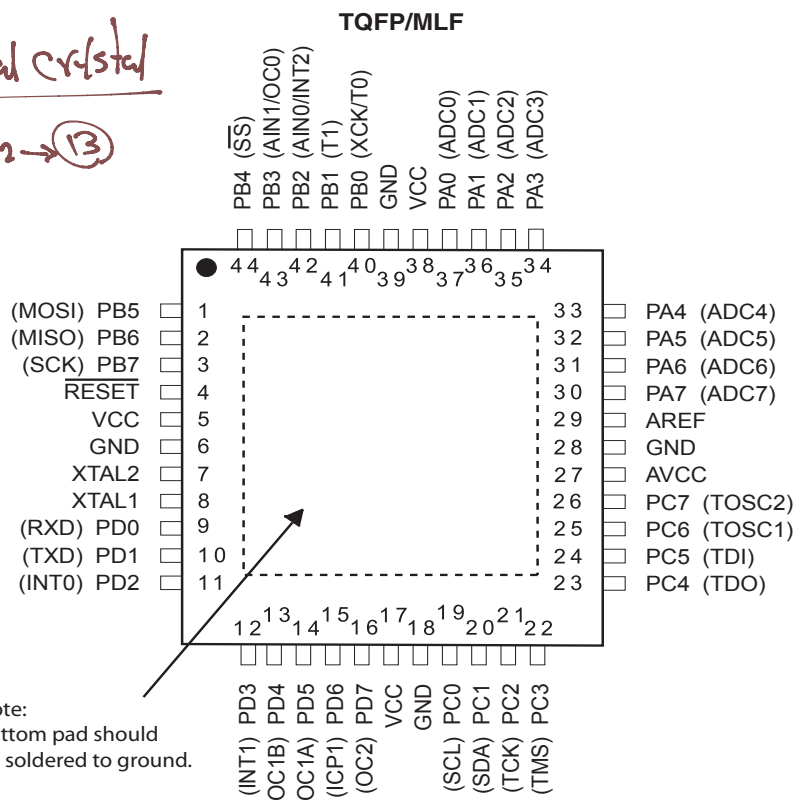
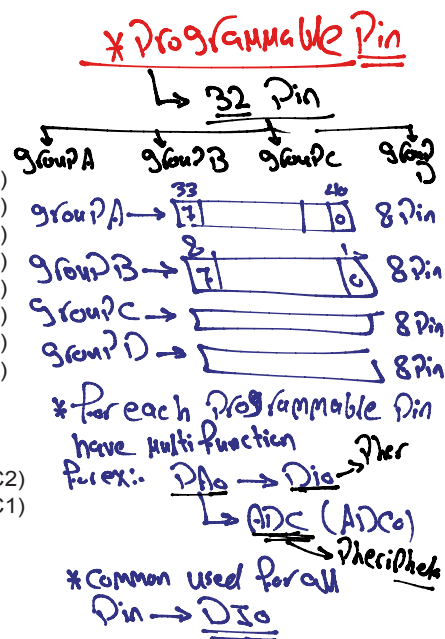
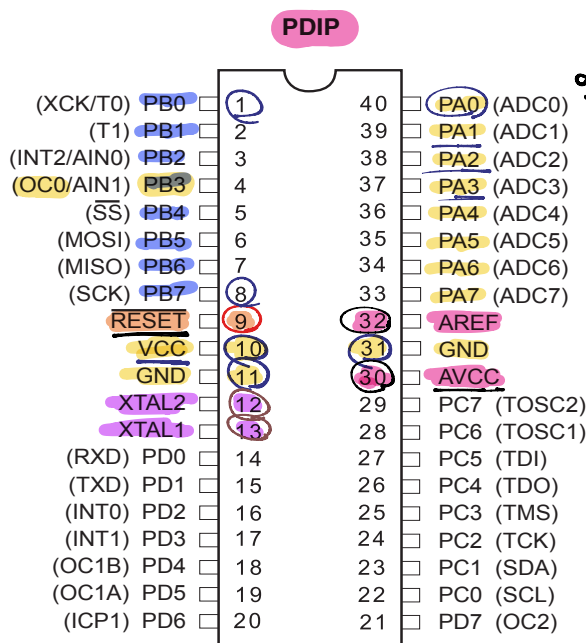
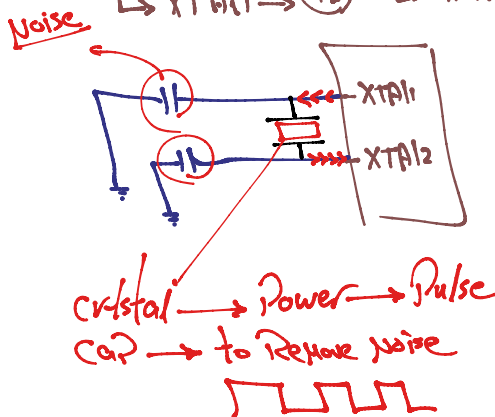
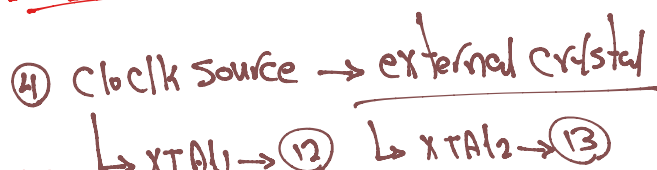
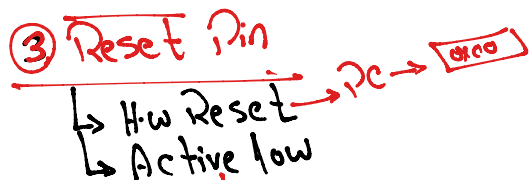
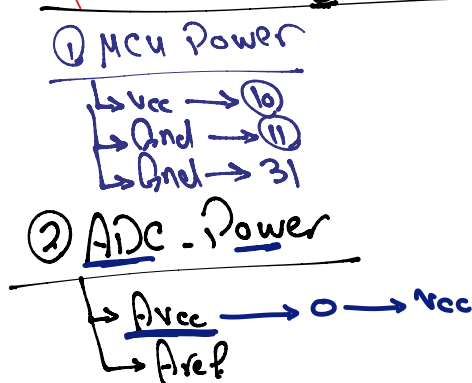
ATmega32
 ATmega32L → low power

Summary

- timer
 - timer → 8 bit (2)
 - timer → 16 bit (1)
 - timer → WDT
- ADC
 - # No → 1
 - channel → 8 channel



(Not Programmable) Pin



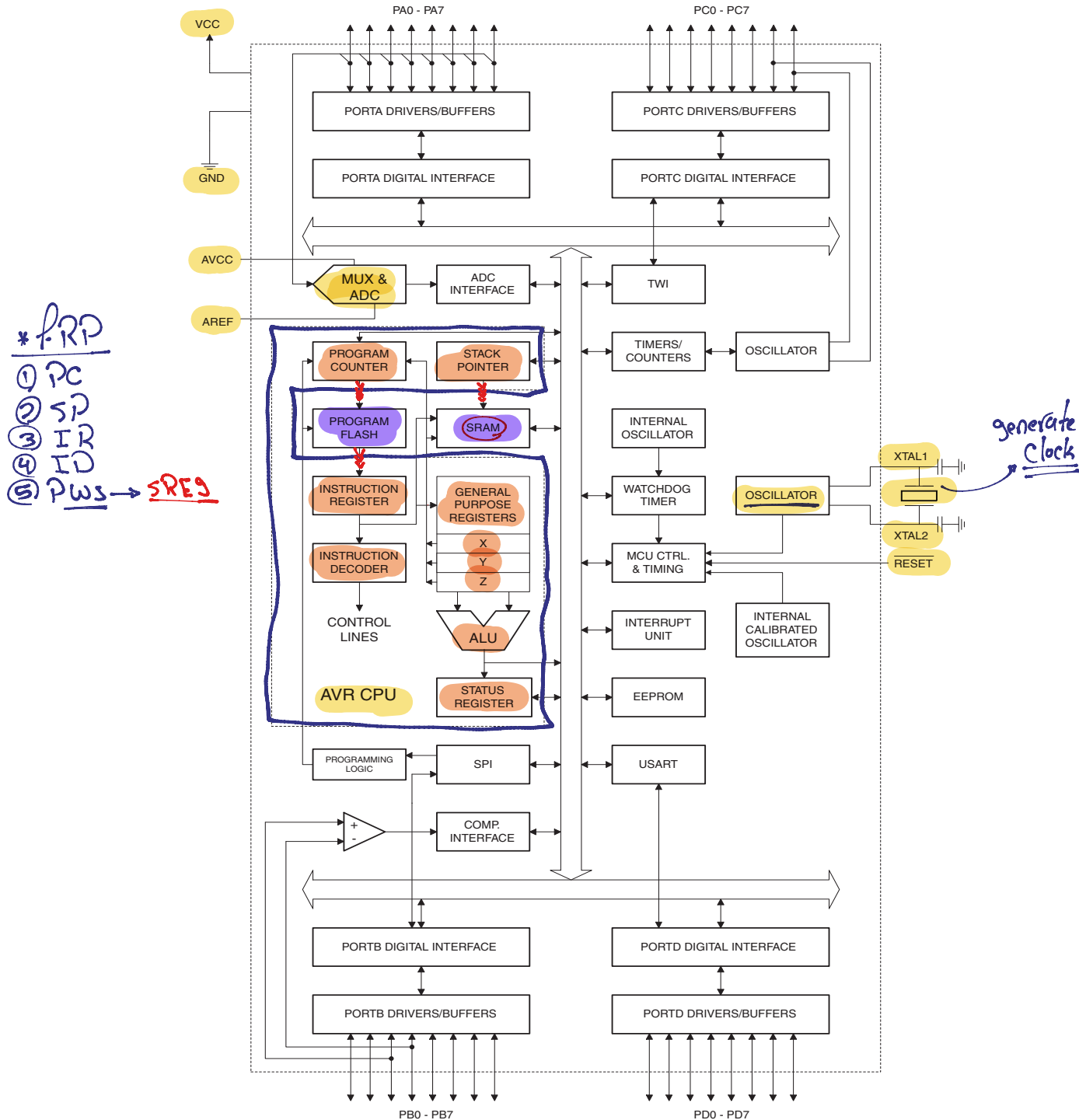
Note:
Bottom pad should
be soldered to ground.

Overview

The Atmel® AVR® ATmega32 is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC architecture. By executing powerful instructions in a single clock cycle, the ATmega32 achieves throughputs approaching 1 MIPS per MHz allowing the system designer to optimize power consumption versus processing speed.

Block Diagram

Figure 2. Block Diagram



The Atmel® AVR® core combines a rich instruction set with 32 general purpose working registers. All the 32 registers are directly connected to the Arithmetic Logic Unit (ALU), allowing two independent registers to be accessed in one single instruction executed in one clock cycle. The resulting architecture is more code efficient while achieving throughputs up to ten times faster than conventional CISC microcontrollers.

The ATmega32 provides the following features: 32Kbytes of In-System Programmable Flash Program memory with Read-While-Write capabilities, 1024bytes EEPROM, 2Kbyte SRAM, 32 general purpose I/O lines, 32 general purpose working registers, a JTAG interface for Boundary-scan, On-chip Debugging support and programming, three flexible Timer/Counters with compare modes, Internal and External Interrupts, a serial programmable USART, a byte oriented Two-wire Serial Interface, an 8-channel, 10-bit ADC with optional differential input stage with programmable gain (TQFP package only), a programmable Watchdog Timer with Internal Oscillator, an SPI serial port, and six software selectable power saving modes. The Idle mode stops the CPU while allowing the USART, Two-wire interface, A/D Converter, SRAM, Timer/Counters, SPI port, and interrupt system to continue functioning. The Power-down mode saves the register contents but freezes the Oscillator, disabling all other chip functions until the next External Interrupt or Hardware Reset. In Power-save mode, the Asynchronous Timer continues to run, allowing the user to maintain a timer base while the rest of the device is sleeping. The ADC Noise Reduction mode stops the CPU and all I/O modules except Asynchronous Timer and ADC, to minimize switching noise during ADC conversions. In Standby mode, the crystal/resonator Oscillator is running while the rest of the device is sleeping. This allows very fast start-up combined with low-power consumption. In Extended Standby mode, both the main Oscillator and the Asynchronous Timer continue to run.

The device is manufactured using Atmel's high density nonvolatile memory technology. The On-chip ISP Flash allows the program memory to be reprogrammed in-system through an SPI serial interface, by a conventional nonvolatile memory programmer, or by an On-chip Boot program running on the AVR core. The boot program can use any interface to download the application program in the Application Flash memory. Software in the Boot Flash section will continue to run while the Application Flash section is updated, providing true Read-While-Write operation. By combining an 8-bit RISC CPU with In-System Self-Programmable Flash on a monolithic chip, the Atmel ATmega32 is a powerful microcontroller that provides a highly-flexible and cost-effective solution to many embedded control applications.

The Atmel AVR ATmega32 is supported with a full suite of program and system development tools including: C compilers, macro assemblers, program debugger/simulators, in-circuit emulators, and evaluation kits.

Pin Descriptions

VCC Digital supply voltage.

GND Ground.

Port A (PA7..PA0) Port A serves as the analog inputs to the A/D Converter.

Port A also serves as an 8-bit bi-directional I/O port, if the A/D Converter is not used. Port pins can provide internal pull-up resistors (selected for each bit). The Port A output buffers have symmetrical drive characteristics with both high sink and source capability. When pins PA0 to PA7 are used as inputs and are externally pulled low, they will source current if the internal pull-up resistors are activated. The Port A pins are tri-stated when a reset condition becomes active, even if the clock is not running.

Port B (PB7..PB0)	Port B is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port B output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port B pins that are externally pulled low will source current if the pull-up resistors are activated. The Port B pins are tri-stated when a reset condition becomes active, even if the clock is not running. Port B also serves the functions of various special features of the ATmega32 as listed on page 57.
Port C (PC7..PC0)	Port C is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port C output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port C pins that are externally pulled low will source current if the pull-up resistors are activated. The Port C pins are tri-stated when a reset condition becomes active, even if the clock is not running. If the JTAG interface is enabled, the pull-up resistors on pins PC5(TDI), PC3(TMS) and PC2(TCK) will be activated even if a reset occurs. The TD0 pin is tri-stated unless TAP states that shift out data are entered. Port C also serves the functions of the JTAG interface and other special features of the ATmega32 as listed on page 60.
Port D (PD7..PD0)	Port D is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port D output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port D pins that are externally pulled low will source current if the pull-up resistors are activated. The Port D pins are tri-stated when a reset condition becomes active, even if the clock is not running. Port D also serves the functions of various special features of the ATmega32 as listed on page 62.
$\overline{\text{RESET}}$	Reset Input. A low level on this pin for longer than the minimum pulse length will generate a reset, even if the clock is not running. The minimum pulse length is given in Table 15 on page 37. Shorter pulses are not guaranteed to generate a reset.
XTAL1	Input to the inverting Oscillator amplifier and input to the internal clock operating circuit.
XTAL2	Output from the inverting Oscillator amplifier.
AVCC	AVCC is the supply voltage pin for Port A and the A/D Converter. It should be externally connected to V_{CC}, even if the ADC is not used. If the ADC is used, it should be connected to V_{CC} through a low-pass filter.
AREF	AREF is the analog reference pin for the A/D Converter.

Resources

A comprehensive set of development tools, application notes and datasheets are available for download on <http://www.atmel.com/avr>.

Data Retention

Reliability Qualification results show that the projected data retention failure rate is much less than 1 PPM over 20 years at 85°C or 100 years at 25°C.

About Code Examples

This documentation contains simple code examples that briefly show how to use various parts of the device. These code examples assume that the part specific header file is included before compilation. Be aware that not all C Compiler vendors include bit definitions in the header files and interrupt handling in C is compiler dependent. Please confirm with the C Compiler documentation for more details.

Register Summary

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Page
\$3F (\$5F)	SREG	I	T	H	S	V	N	Z	C	10
\$3E (\$5E)	SPH	—	—	—	—	SP11	SP10	SP9	SP8	12
\$3D (\$5D)	SPL	SP7	SP6	SP5	SP4	SP3	SP2	SP1	SP0	12
\$3C (\$5C)	OCR0	Timer/Counter0 Output Compare Register								82
\$3B (\$5B)	GICR	INT1	INT0	INT2	—	—	—	IVSEL	IVCE	47, 67
\$3A (\$5A)	GIFR	INTF1	INTF0	INTF2	—	—	—	—	—	68
\$39 (\$59)	TIMSK	OCIE2	TOIE2	TICIE1	OCIE1A	OCIE1B	TOIE1	OCIE0	TOIE0	82, 112, 130
\$38 (\$58)	TIFR	OCF2	TOV2	ICF1	OCF1A	OCF1B	TOV1	OCF0	TOV0	83, 112, 130
\$37 (\$57)	SPMCR	SPMIE	RWWWSB	—	RWWWSRE	BLBSET	PGWRT	PGERS	SPMEN	248
\$36 (\$56)	TWCR	TWINT	TWEA	TWSTA	TWSTO	TWWC	TWEN	—	TWIE	177
\$35 (\$55)	MCUCR	SE	SM2	SM1	SM0	ISC11	ISC10	ISC01	ISC00	32, 66
\$34 (\$54)	MCUCSR	JTD	ISC2	—	JTRF	WDRF	BORF	EXTRF	PORF	40, 67, 228
\$33 (\$53)	TCCR0	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00	80
\$32 (\$52)	TCNT0	Timer/Counter0 (8 Bits)								82
\$31 ⁽¹⁾ (\$51) ⁽¹⁾	OSCCAL	Oscillator Calibration Register								30
	ODCR	On-Chip Debug Register								224
\$30 (\$50)	SFIOR	ADTS2	ADTS1	ADTS0	—	ACME	PUD	PSR2	PSR10	56, 85, 131, 198, 218
\$2F (\$4F)	TCCR1A	COM1A1	COM1A0	COM1B1	COM1B0	FOC1A	FOC1B	WGM11	WGM10	107
\$2E (\$4E)	TCCR1B	ICNC1	ICES1	—	WGM13	WGM12	CS12	CS11	CS10	110
\$2D (\$4D)	TCNT1H	Timer/Counter1 – Counter Register High Byte								111
\$2C (\$4C)	TCNT1L	Timer/Counter1 – Counter Register Low Byte								111
\$2B (\$4B)	OCR1AH	Timer/Counter1 – Output Compare Register A High Byte								111
\$2A (\$4A)	OCR1AL	Timer/Counter1 – Output Compare Register A Low Byte								111
\$29 (\$49)	OCR1BH	Timer/Counter1 – Output Compare Register B High Byte								111
\$28 (\$48)	OCR1BL	Timer/Counter1 – Output Compare Register B Low Byte								111
\$27 (\$47)	ICR1H	Timer/Counter1 – Input Capture Register High Byte								111
\$26 (\$46)	ICR1L	Timer/Counter1 – Input Capture Register Low Byte								111
\$25 (\$45)	TCCR2	FOC2	WGM20	COM21	COM20	WGM21	CS22	CS21	CS20	125
\$24 (\$44)	TCNT2	Timer/Counter2 (8 Bits)								127
\$23 (\$43)	OCR2	Timer/Counter2 Output Compare Register								127
\$22 (\$42)	ASSR	—	—	—	—	AS2	TCN2UB	OCR2UB	TCR2UB	128
\$21 (\$41)	WDTCSR	—	—	—	WDTOE	WDE	WDP2	WDP1	WDP0	42
\$20 ⁽²⁾ (\$40) ⁽²⁾	UBRRH	URSEL	—	—	—	UBRR[11:8]				164
	UCSRC	URSEL	UMSEL	UPM1	UPM0	USBS	UCSZ1	UCSZ0	UCPOL	162
\$1F (\$3F)	EEARH	—	—	—	—	—	—	EEAR9	EEAR8	19
\$1E (\$3E)	EEARL	EEPROM Address Register Low Byte								19
\$1D (\$3D)	EEDR	EEPROM Data Register								19
\$1C (\$3C)	EEDR	—	—	—	—	EERIE	EEMWE	EERE	EERE	19
\$1B (\$3B)	PORTA	PORTA7	PORTA6	PORTA5	PORTA4	PORTA3	PORTA2	PORTA1	PORTA0	64
\$1A (\$3A)	DDRA	DDA7	DDA6	DDA5	DDA4	DDA3	DDA2	DDA1	DDA0	64
\$19 (\$39)	PINA	PINA7	PINA6	PINA5	PINA4	PINA3	PINA2	PINA1	PINA0	64
\$18 (\$38)	PORTB	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	64
\$17 (\$37)	DDRB	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0	64
\$16 (\$36)	PINB	PINB7	PINB6	PINB5	PINB4	PINB3	PINB2	PINB1	PINB0	65
\$15 (\$35)	PORTC	PORTC7	PORTC6	PORTC5	PORTC4	PORTC3	PORTC2	PORTC1	PORTC0	65
\$14 (\$34)	DDRC	DDC7	DDC6	DDC5	DDC4	DDC3	DDC2	DDC1	DDC0	65
\$13 (\$33)	PINC	PINC7	PINC6	PINC5	PINC4	PINC3	PINC2	PINC1	PINC0	65
\$12 (\$32)	PORTD	PORTD7	PORTD6	PORTD5	PORTD4	PORTD3	PORTD2	PORTD1	PORTD0	65
\$11 (\$31)	DDRD	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	65
\$10 (\$30)	PIND	PIND7	PIND6	PIND5	PIND4	PIND3	PIND2	PIND1	PIND0	65
\$0F (\$2F)	SPDR	SPI Data Register								138
\$0E (\$2E)	SPSR	SPIF	WCOL	—	—	—	—	—	SPI2X	138
\$0D (\$2D)	SPCR	SPIE	SPE	DORD	MSTR	CPOL	CPHA	SPR1	SPR0	136
\$0C (\$2C)	UDR	USART I/O Data Register								159
\$0B (\$2B)	UCSRA	RXC	TXC	UDRE	FE	DOR	PE	U2X	MPCM	160
\$0A (\$2A)	UCSRB	RXCIE	TXCIE	UDRIE	RXEN	TXEN	UCSZ2	RXB8	TXB8	161
\$09 (\$29)	UBRRL	USART Baud Rate Register Low Byte								164
\$08 (\$28)	ACSR	ACD	ACBG	ACO	ACI	ACIE	ACIC	ACIS1	ACIS0	199
\$07 (\$27)	ADMUX	REFS1	REFS0	ADLAR	MUX4	MUX3	MUX2	MUX1	MUX0	214
\$06 (\$26)	ADCSRA	ADEN	ADSC	ADATE	ADIF	ADIE	ADPS2	ADPS1	ADPS0	216
\$05 (\$25)	ADCH	ADC Data Register High Byte								217
\$04 (\$24)	ADCL	ADC Data Register Low Byte								217
\$03 (\$23)	TWDR	Two-wire Serial Interface Data Register								179
\$02 (\$22)	TWAR	TWA6	TWA5	TWA4	TWA3	TWA2	TWA1	TWA0	TWGCE	179

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Page
\$01 (\$21)	TWSR	TWS7	TWS6	TWS5	TWS4	TWS3	–	TWPS1	TWPS0	178
\$00 (\$20)	TWBR	Two-wire Serial Interface Bit Rate Register								177

- Notes:
1. When the OCDEN Fuse is unprogrammed, the OSCCAL Register is always accessed on this address. Refer to the debugger specific documentation for details on how to use the OCDR Register.
 2. Refer to the USART description for details on how to access UBRRH and UCSRC.
 3. For compatibility with future devices, reserved bits should be written to zero if accessed. Reserved I/O memory addresses should never be written.
 4. Some of the Status Flags are cleared by writing a logical one to them. Note that the CBI and SBI instructions will operate on all bits in the I/O Register, writing a one back into any flag read as set, thus clearing the flag. The CBI and SBI instructions work with registers \$00 to \$1F only.

Instruction Set Summary

Mnemonics	Operands	Description	Operation	Flags	#Clocks
ARITHMETIC AND LOGIC INSTRUCTIONS					
ADD	Rd, Rr	Add two Registers	$Rd \leftarrow Rd + Rr$	Z,C,N,V,H	1
ADC	Rd, Rr	Add with Carry two Registers	$Rd \leftarrow Rd + Rr + C$	Z,C,N,V,H	1
ADIW	RdI, K	Add Immediate to Word	$Rdh:Rdl \leftarrow Rdh:Rdl + K$	Z,C,N,V,S	2
SUB	Rd, Rr	Subtract two Registers	$Rd \leftarrow Rd - Rr$	Z,C,N,V,H	1
SUBI	Rd, K	Subtract Constant from Register	$Rd \leftarrow Rd - K$	Z,C,N,V,H	1
SBC	Rd, Rr	Subtract with Carry two Registers	$Rd \leftarrow Rd - Rr - C$	Z,C,N,V,H	1
SBCI	Rd, K	Subtract with Carry Constant from Reg.	$Rd \leftarrow Rd - K - C$	Z,C,N,V,H	1
SBIW	RdI, K	Subtract Immediate from Word	$Rdh:Rdl \leftarrow Rdh:Rdl - K$	Z,C,N,V,S	2
AND	Rd, Rr	Logical AND Registers	$Rd \leftarrow Rd \bullet Rr$	Z,N,V	1
ANDI	Rd, K	Logical AND Register and Constant	$Rd \leftarrow Rd \bullet K$	Z,N,V	1
OR	Rd, Rr	Logical OR Registers	$Rd \leftarrow Rd \vee Rr$	Z,N,V	1
ORI	Rd, K	Logical OR Register and Constant	$Rd \leftarrow Rd \vee K$	Z,N,V	1
EOR	Rd, Rr	Exclusive OR Registers	$Rd \leftarrow Rd \oplus Rr$	Z,N,V	1
COM	Rd	One's Complement	$Rd \leftarrow \$FF - Rd$	Z,C,N,V	1
NEG	Rd	Two's Complement	$Rd \leftarrow \$00 - Rd$	Z,C,N,V,H	1
SBR	Rd, K	Set Bit(s) in Register	$Rd \leftarrow Rd \vee K$	Z,N,V	1
CBR	Rd, K	Clear Bit(s) in Register	$Rd \leftarrow Rd \bullet (\$FF - K)$	Z,N,V	1
INC	Rd	Increment	$Rd \leftarrow Rd + 1$	Z,N,V	1
DEC	Rd	Decrement	$Rd \leftarrow Rd - 1$	Z,N,V	1
TST	Rd	Test for Zero or Minus	$Rd \leftarrow Rd \bullet Rd$	Z,N,V	1
CLR	Rd	Clear Register	$Rd \leftarrow Rd \oplus Rd$	Z,N,V	1
SER	Rd	Set Register	$Rd \leftarrow \$FF$	None	1
MUL	Rd, Rr	Multiply Unsigned	$R1:R0 \leftarrow Rd \times Rr$	Z,C	2
MULS	Rd, Rr	Multiply Signed	$R1:R0 \leftarrow Rd \times Rr$	Z,C	2
MULSU	Rd, Rr	Multiply Signed with Unsigned	$R1:R0 \leftarrow Rd \times Rr$	Z,C	2
FMUL	Rd, Rr	Fractional Multiply Unsigned	$R1:R0 \leftarrow (Rd \times Rr) \ll 1$	Z,C	2
FMULS	Rd, Rr	Fractional Multiply Signed	$R1:R0 \leftarrow (Rd \times Rr) \ll 1$	Z,C	2
FMULSU	Rd, Rr	Fractional Multiply Signed with Unsigned	$R1:R0 \leftarrow (Rd \times Rr) \ll 1$	Z,C	2
BRANCH INSTRUCTIONS					
RJMP	k	Relative Jump	$PC \leftarrow PC + k + 1$	None	2
IJMP		Indirect Jump to (Z)	$PC \leftarrow Z$	None	2
JMP	k	Direct Jump	$PC \leftarrow k$	None	3
RCALL	k	Relative Subroutine Call	$PC \leftarrow PC + k + 1$	None	3
ICALL		Indirect Call to (Z)	$PC \leftarrow Z$	None	3
CALL	k	Direct Subroutine Call	$PC \leftarrow k$	None	4
RET		Subroutine Return	$PC \leftarrow \text{Stack}$	None	4
RETI		Interrupt Return	$PC \leftarrow \text{Stack}$	I	4
CPSE	Rd, Rr	Compare, Skip if Equal	if $(Rd = Rr)$ $PC \leftarrow PC + 2$ or 3	None	1 / 2 / 3
CP	Rd, Rr	Compare	$Rd - Rr$	Z, N, V, C, H	1
CPC	Rd, Rr	Compare with Carry	$Rd - Rr - C$	Z, N, V, C, H	1
CPI	Rd, K	Compare Register with Immediate	$Rd - K$	Z, N, V, C, H	1
SBRC	Rr, b	Skip if Bit in Register Cleared	if $(Rr(b)=0)$ $PC \leftarrow PC + 2$ or 3	None	1 / 2 / 3
SBRS	Rr, b	Skip if Bit in Register is Set	if $(Rr(b)=1)$ $PC \leftarrow PC + 2$ or 3	None	1 / 2 / 3
SBIC	P, b	Skip if Bit in I/O Register Cleared	if $(P(b)=0)$ $PC \leftarrow PC + 2$ or 3	None	1 / 2 / 3
SBIS	P, b	Skip if Bit in I/O Register is Set	if $(P(b)=1)$ $PC \leftarrow PC + 2$ or 3	None	1 / 2 / 3
BRBS	s, k	Branch if Status Flag Set	if $(SREG(s) = 1)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRBC	s, k	Branch if Status Flag Cleared	if $(SREG(s) = 0)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BREQ	k	Branch if Equal	if $(Z = 1)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRNE	k	Branch if Not Equal	if $(Z = 0)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRCS	k	Branch if Carry Set	if $(C = 1)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRCC	k	Branch if Carry Cleared	if $(C = 0)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRSH	k	Branch if Same or Higher	if $(C = 0)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRLO	k	Branch if Lower	if $(C = 1)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRMI	k	Branch if Minus	if $(N = 1)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRPL	k	Branch if Plus	if $(N = 0)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRGE	k	Branch if Greater or Equal, Signed	if $(N \oplus V = 0)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRLT	k	Branch if Less Than Zero, Signed	if $(N \oplus V = 1)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRHS	k	Branch if Half Carry Flag Set	if $(H = 1)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRHC	k	Branch if Half Carry Flag Cleared	if $(H = 0)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRTS	k	Branch if T Flag Set	if $(T = 1)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRTC	k	Branch if T Flag Cleared	if $(T = 0)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRVS	k	Branch if Overflow Flag is Set	if $(V = 1)$ then $PC \leftarrow PC + k + 1$	None	1 / 2
BRVC	k	Branch if Overflow Flag is Cleared	if $(V = 0)$ then $PC \leftarrow PC + k + 1$	None	1 / 2

Mnemonics	Operands	Description	Operation	Flags	#Clocks
BRIE	k	Branch if Interrupt Enabled	if (I = 1) then PC ← PC + k + 1	None	1 / 2
BRID	k	Branch if Interrupt Disabled	if (I = 0) then PC ← PC + k + 1	None	1 / 2
DATA TRANSFER INSTRUCTIONS					
MOV	Rd, Rr	Move Between Registers	Rd ← Rr	None	1
MOVW	Rd, Rr	Copy Register Word	Rd+1:Rd ← Rr+1:Rr	None	1
LDI	Rd, K	Load Immediate	Rd ← K	None	1
LD	Rd, X	Load Indirect	Rd ← (X)	None	2
LD	Rd, X+	Load Indirect and Post-Inc.	Rd ← (X), X ← X + 1	None	2
LD	Rd, -X	Load Indirect and Pre-Dec.	X ← X - 1, Rd ← (X)	None	2
LD	Rd, Y	Load Indirect	Rd ← (Y)	None	2
LD	Rd, Y+	Load Indirect and Post-Inc.	Rd ← (Y), Y ← Y + 1	None	2
LD	Rd, -Y	Load Indirect and Pre-Dec.	Y ← Y - 1, Rd ← (Y)	None	2
LDD	Rd, Y+q	Load Indirect with Displacement	Rd ← (Y + q)	None	2
LD	Rd, Z	Load Indirect	Rd ← (Z)	None	2
LD	Rd, Z+	Load Indirect and Post-Inc.	Rd ← (Z), Z ← Z + 1	None	2
LD	Rd, -Z	Load Indirect and Pre-Dec.	Z ← Z - 1, Rd ← (Z)	None	2
LDD	Rd, Z+q	Load Indirect with Displacement	Rd ← (Z + q)	None	2
LDS	Rd, k	Load Direct from SRAM	Rd ← (k)	None	2
ST	X, Rr	Store Indirect	(X) ← Rr	None	2
ST	X+, Rr	Store Indirect and Post-Inc.	(X) ← Rr, X ← X + 1	None	2
ST	-X, Rr	Store Indirect and Pre-Dec.	X ← X - 1, (X) ← Rr	None	2
ST	Y, Rr	Store Indirect	(Y) ← Rr	None	2
ST	Y+, Rr	Store Indirect and Post-Inc.	(Y) ← Rr, Y ← Y + 1	None	2
ST	-Y, Rr	Store Indirect and Pre-Dec.	Y ← Y - 1, (Y) ← Rr	None	2
STD	Y+q, Rr	Store Indirect with Displacement	(Y + q) ← Rr	None	2
ST	Z, Rr	Store Indirect	(Z) ← Rr	None	2
ST	Z+, Rr	Store Indirect and Post-Inc.	(Z) ← Rr, Z ← Z + 1	None	2
ST	-Z, Rr	Store Indirect and Pre-Dec.	Z ← Z - 1, (Z) ← Rr	None	2
STD	Z+q, Rr	Store Indirect with Displacement	(Z + q) ← Rr	None	2
STS	k, Rr	Store Direct to SRAM	(k) ← Rr	None	2
LPM		Load Program Memory	R0 ← (Z)	None	3
LPM	Rd, Z	Load Program Memory	Rd ← (Z)	None	3
LPM	Rd, Z+	Load Program Memory and Post-Inc	Rd ← (Z), Z ← Z + 1	None	3
SPM		Store Program Memory	(Z) ← R1:R0	None	-
IN	Rd, P	In Port	Rd ← P	None	1
OUT	P, Rr	Out Port	P ← Rr	None	1
PUSH	Rr	Push Register on Stack	Stack ← Rr	None	2
POP	Rd	Pop Register from Stack	Rd ← Stack	None	2
BIT AND BIT-TEST INSTRUCTIONS					
SBI	P, b	Set Bit in I/O Register	I/O(P, b) ← 1	None	2
CBI	P, b	Clear Bit in I/O Register	I/O(P, b) ← 0	None	2
LSL	Rd	Logical Shift Left	Rd(n+1) ← Rd(n), Rd(0) ← 0	Z, C, N, V	1
LSR	Rd	Logical Shift Right	Rd(n) ← Rd(n+1), Rd(7) ← 0	Z, C, N, V	1
ROL	Rd	Rotate Left Through Carry	Rd(0) ← C, Rd(n+1) ← Rd(n), C ← Rd(7)	Z, C, N, V	1
ROR	Rd	Rotate Right Through Carry	Rd(7) ← C, Rd(n) ← Rd(n+1), C ← Rd(0)	Z, C, N, V	1
ASR	Rd	Arithmetic Shift Right	Rd(n) ← Rd(n+1), n=0..6	Z, C, N, V	1
SWAP	Rd	Swap Nibbles	Rd(3..0) ← Rd(7..4), Rd(7..4) ← Rd(3..0)	None	1
BSET	s	Flag Set	SREG(s) ← 1	SREG(s)	1
BCLR	s	Flag Clear	SREG(s) ← 0	SREG(s)	1
BST	Rr, b	Bit Store from Register to T	T ← Rr(b)	T	1
BLD	Rd, b	Bit load from T to Register	Rd(b) ← T	None	1
SEC		Set Carry	C ← 1	C	1
CLC		Clear Carry	C ← 0	C	1
SEN		Set Negative Flag	N ← 1	N	1
CLN		Clear Negative Flag	N ← 0	N	1
SEZ		Set Zero Flag	Z ← 1	Z	1
CLZ		Clear Zero Flag	Z ← 0	Z	1
SEI		Global Interrupt Enable	I ← 1	I	1
CLI		Global Interrupt Disable	I ← 0	I	1
SES		Set Signed Test Flag	S ← 1	S	1
CLS		Clear Signed Test Flag	S ← 0	S	1
SEV		Set Twos Complement Overflow	V ← 1	V	1
CLV		Clear Twos Complement Overflow	V ← 0	V	1
SET		Set T in SREG	T ← 1	T	1
CLT		Clear T in SREG	T ← 0	T	1
SEH		Set Half Carry Flag in SREG	H ← 1	H	1

Mnemonics	Operands	Description	Operation	Flags	#Clocks
CLH		Clear Half Carry Flag in SREG	$H \leftarrow 0$	H	1
MCU CONTROL INSTRUCTIONS					
NOP		No Operation		None	1
SLEEP		Sleep	(see specific descr. for Sleep function)	None	1
WDR		Watchdog Reset	(see specific descr. for WDR/timer)	None	1
BREAK		Break	For On-Chip Debug Only	None	N/A

Ordering Information

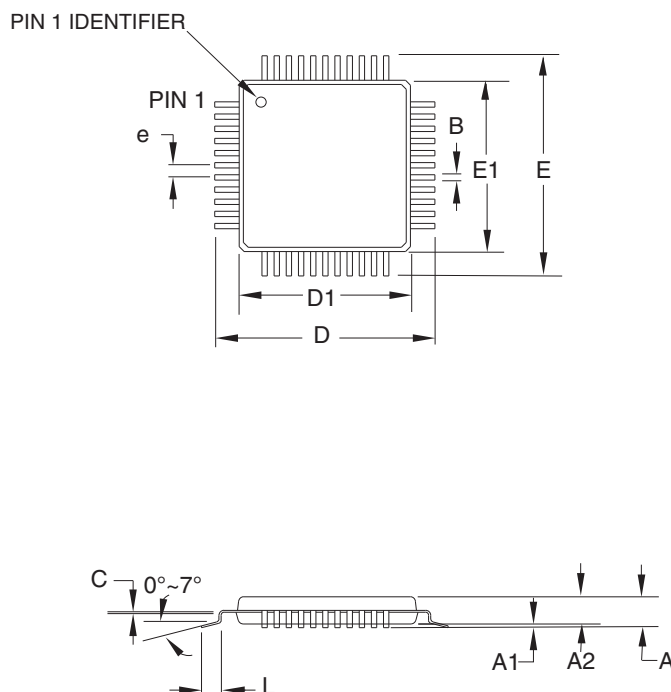
Speed (MHz)	Power Supply	Ordering Code ⁽²⁾	Package ⁽¹⁾	Operational Range
8	2.7V - 5.5V	ATmega32L-8AU	44A	Industrial (-40°C to 85°C)
		ATmega32L-8AUR ⁽³⁾	44A	
		ATmega32L-8PU	40P6	
		ATmega32L-8MU	44M1	
		ATmega32L-8MUR ⁽³⁾	44M1	
16	4.5V - 5.5V	ATmega32-16AU	44A	
		ATmega32-16AUR ⁽³⁾	44A	
		ATmega32-16PU	40P6	
		ATmega32-16MU	44M1	
		ATmega32-16MUR ⁽³⁾	44M1	

- Notes:
1. This device can also be supplied in wafer form. Please contact your local Atmel sales office for detailed ordering information and minimum quantities.
 2. Pb-free packaging complies to the European Directive for Restriction of Hazardous Substances (RoHS directive). Also Halide free and fully Green.
 3. Tape & Reel

Package Type	
44A	44-lead, 10 × 10 × 1.0mm, Thin Profile Plastic Quad Flat Package (TQFP)
40P6	40-pin, 0.600" Wide, Plastic Dual Inline Package (PDIP)
44M1	44-pad, 7 × 7 × 1.0mm, Quad Flat No-Lead/Micro Lead Frame Package (QFN/MLF)

Packaging Information

44A



COMMON DIMENSIONS
(Unit of Measure = mm)

SYMBOL	MIN	NOM	MAX	NOTE
A	—	—	1.20	
A1	0.05	—	0.15	
A2	0.95	1.00	1.05	
D	11.75	12.00	12.25	
D1	9.90	10.00	10.10	Note 2
E	11.75	12.00	12.25	
E1	9.90	10.00	10.10	Note 2
B	0.30	—	0.45	
C	0.09	—	0.20	
L	0.45	—	0.75	
e	0.80 TYP			

Notes:

1. This package conforms to JEDEC reference MS-026, Variation ACB.
2. Dimensions D1 and E1 do not include mold protrusion. Allowable protrusion is 0.25 mm per side. Dimensions D1 and E1 are maximum plastic body size dimensions including mold mismatch.
3. Lead coplanarity is 0.10 mm maximum.

2010-10-20



2325 Orchard Parkway
San Jose, CA 95131

TITLE

44A, 44-lead, 10 x 10 mm Body Size, 1.0 mm Body Thickness,
0.8 mm Lead Pitch, Thin Profile Plastic Quad Flat Package (TQFP)

DRAWING NO.

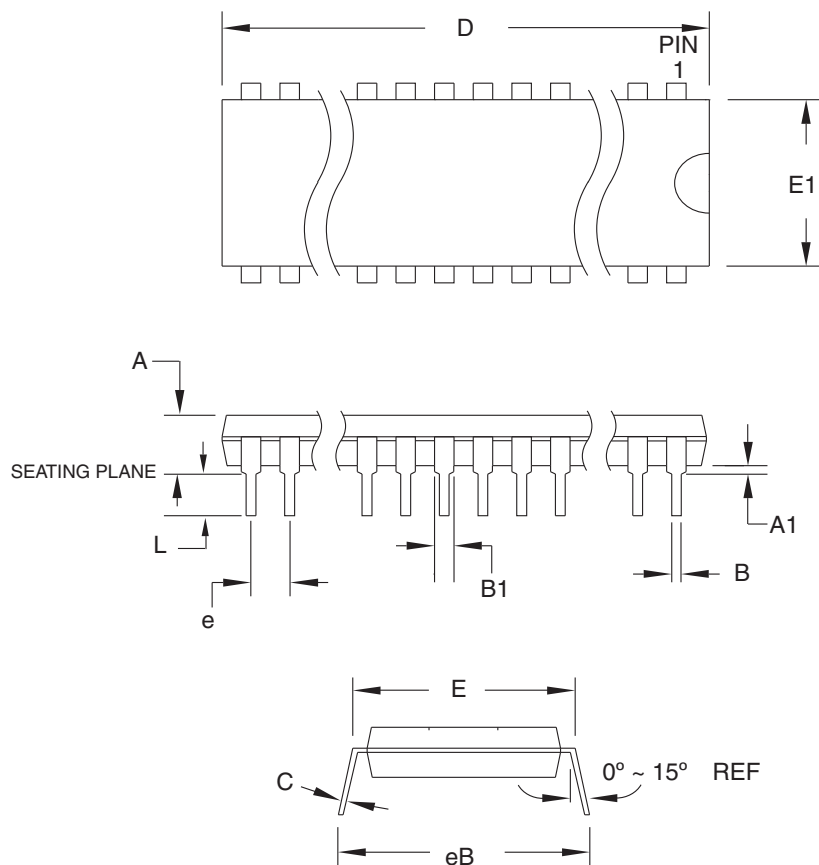
44A

REV.

C



40P6



COMMON DIMENSIONS
(Unit of Measure = mm)

SYMBOL	MIN	NOM	MAX	NOTE
A	—	—	4.826	
A1	0.381	—	—	
D	52.070	—	52.578	Note 2
E	15.240	—	15.875	
E1	13.462	—	13.970	Note 2
B	0.356	—	0.559	
B1	1.041	—	1.651	
L	3.048	—	3.556	
C	0.203	—	0.381	
eB	15.494	—	17.526	
e	2.540 TYP			

- Notes:
1. This package conforms to JEDEC reference MS-011, Variation AC.
 2. Dimensions D and E1 do not include mold Flash or Protrusion. Mold Flash or Protrusion shall not exceed 0.25 mm (0.010").

09/28/01



2325 Orchard Parkway
San Jose, CA 95131

TITLE

40P6, 40-lead (0.600"/15.24 mm Wide) Plastic Dual
Inline Package (PDIP)

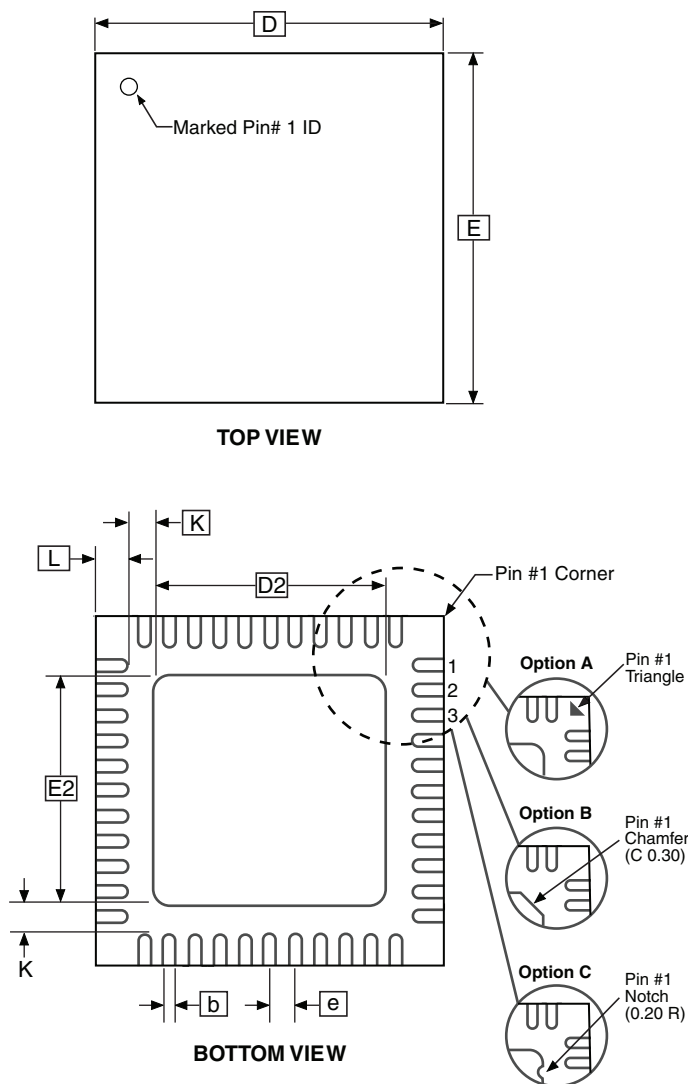
DRAWING NO.

40P6

REV.

B

44M1



COMMON DIMENSIONS
(Unit of Measure = mm)

SYMBOL	MIN	NOM	MAX	NOTE
A	0.80	0.90	1.00	
A1	—	0.02	0.05	
A3	0.20 REF			
b	0.18	0.23	0.30	
D	6.90	7.00	7.10	
D2	5.00	5.20	5.40	
E	6.90	7.00	7.10	
E2	5.00	5.20	5.40	
e	0.50 BSC			
L	0.59	0.64	0.69	
K	0.20	0.26	0.41	

Note: JEDEC Standard MO-220, Fig. 1 (SAW Singulation) VKKD-3.

9/26/08



Package Drawing Contact:
packagedrawings@atmel.com

TITLE
44M1, 44-pad, 7 x 7 x 1.0 mm Body, Lead Pitch 0.50 mm, 5.20 mm Exposed Pad, Thermally Enhanced Plastic Very Thin Quad Flat No Lead Package (VQFN)

GPC

ZWS

DRAWING NO.

44M1

REV.

H

Errata

ATmega32, rev. A to F

- First Analog Comparator conversion may be delayed
- Interrupts may be lost when writing the timer registers in the asynchronous timer
- IDCODE masks data from TDI input
- Reading EEPROM by using ST or STS to set EERE bit triggers unexpected interrupt request.

1. First Analog Comparator conversion may be delayed

If the device is powered by a slow rising V_{CC} , the first Analog Comparator conversion will take longer than expected on some devices.

Problem Fix/Workaround

When the device has been powered or reset, disable then enable the Analog Comparator before the first conversion.

2. Interrupts may be lost when writing the timer registers in the asynchronous timer

The interrupt will be lost if a timer register that is synchronous timer clock is written when the asynchronous Timer/Counter register (TCNTx) is 0x00.

Problem Fix/Workaround

Always check that the asynchronous Timer/Counter register neither have the value 0xFF nor 0x00 before writing to the asynchronous Timer Control Register (TCCRx), asynchronous Timer Counter Register (TCNTx), or asynchronous Output Compare Register (OCRx).

3. IDCODE masks data from TDI input

The JTAG instruction IDCODE is not working correctly. Data to succeeding devices are replaced by all-ones during Update-DR.

Problem Fix / Workaround

- If ATmega32 is the only device in the scan chain, the problem is not visible.
- Select the Device ID Register of the ATmega32 by issuing the IDCODE instruction or by entering the Test-Logic-Reset state of the TAP controller to read out the contents of its Device ID Register and possibly data from succeeding devices of the scan chain. Issue the BYPASS instruction to the ATmega32 while reading the Device ID Registers of preceding devices of the boundary scan chain.
- If the Device IDs of all devices in the boundary scan chain must be captured simultaneously, the ATmega32 must be the first device in the chain.

4. Reading EEPROM by using ST or STS to set EERE bit triggers unexpected interrupt request.

Reading EEPROM by using the ST or STS command to set the EERE bit in the EECR register triggers an unexpected EEPROM interrupt request.

Problem Fix / Workaround

Always use OUT or SBI to set EERE in EECR.

Datasheet Revision History

Please note that the referring page numbers in this section are referred to this document. The referring revision in this section are referring to the document revision.

Changes from Rev. 2503P-07/09 to Rev. 2503Q-02/11

1. Updated **“Packaging Information” on page 333**, by replacing the package 44A by a correct one.
2. Updated the datasheet according to the Atmel new Brand Style Guide.
3. Updated **“Ordering Information” on page 332** to include Tape & Reel devices.

Changes from Rev. 2503O-07/09 to Rev. 2503P-07/10

1. Inserted Note in **“Performing Page Erase by SPM” on page 251**.
2. Note 6 and Note 7 in **Table 119 on page 290** have been removed.
3. Updated **“Performing Page Erase by SPM” on page 251**.

Changes from Rev. 2503N-06/08 to Rev. 2503O-07/09

1. Updated **“Errata” on page 336**.
2. Updated the TOC with new template (version 5.10)

Changes from Rev. 2503M-05/08 to Rev. 2503N-06/08

1. Added the note **“Not recommended for new designs” on “Features” on page 1**.

Changes from Rev. 2503L-05/08 to Rev. 2503M-05/08

1. Updated **“Ordering Information” on page 12**:
 - Commercial ordering codes removed.
 - Non Pb-free package option removed.
2. Removed note from Feature list in **“Analog to Digital Converter” on page 201**.
3. Removed note from **Table 84 on page 215**.

Changes from Rev. 2503K-08/07 to Rev. 2503L-05/08

1. Updated **“Fast PWM Mode” on page 75** in **“8-bit Timer/Counter0 with PWM” on page 69**:
 - Removed the last section describing how to achieve a frequency with 50% duty cycle waveform output in fast PWM mode.

Changes from Rev. 2503J-10/06 to Rev. 2503K-08/07

1. Renamed **“Input Capture Trigger Source” to “Input Capture Pin Source” on page 94**.
2. Updated **“Features” on page 1**.
3. Added **“Data Retention” on page 6**.
4. Updated **“Errata” on page 336**.

5. Updated “Slave Mode” on page 136.

Changes from Rev. 2503I-04/06 to Rev. 2503J-10/06

1. Updated “Fast PWM Mode” on page 99.
2. Updated Table 38 on page 80, Table 40 on page 81, Table 45 on page 108, Table 47 on page 109, Table 50 on page 125 and Table 52 on page 126.
3. Updated typo in table note 6 in “DC Characteristics” on page 287.
4. Updated “Errata” on page 336.

Changes from Rev. 2503H-03/05 to Rev. 2503I-04/06

1. Updated Figure 1 on page 2.
2. Added “Resources” on page 6.
3. Added note to “Timer/Counter Oscillator” on page 31.
4. Updated “Serial Peripheral Interface – SPI” on page 132.
5. Updated note in “Bit Rate Generator Unit” on page 175.
6. Updated Table 86 on page 218.
7. Updated “DC Characteristics” on page 287.

Changes from Rev. 2503G-11/04 to Rev. 2503H-03/05

1. MLF-package alternative changed to “Quad Flat No-Lead/Micro Lead Frame Package QFN/MLF”.
2. Updated “Electrical Characteristics” on page 287
3. Updated “Ordering Information” on page 332.

Changes from Rev. 2503F-12/03 to Rev. 2503G-11/04

1. “Channel” renamed “Compare unit” in Timer/Counter sections, ICP renamed ICP1.
2. Updated Table 7 on page 29, Table 15 on page 37, Table 81 on page 206, Table 114 on page 272, Table 115 on page 273, and Table 118 on page 289.
3. Updated Figure 1 on page 2, Figure 46 on page 100.
4. Updated “Version” on page 226.
5. Updated “Calibration Byte” on page 258.
6. Added section “Page Size” on page 258.
7. Updated “ATmega32 Typical Characteristics” on page 296.
8. Updated “Ordering Information” on page 332.

Changes from Rev. 2503E-09/03 to Rev. 2503F-12/03

1. Updated “Calibrated Internal RC Oscillator” on page 29.

Changes from Rev. 2503D-02/03 to Rev. 2503E-09/03

1. Updated and changed “On-chip Debug System” to “[JTAG Interface and On-chip Debug System](#)” on page 35.
2. Updated [Table 15](#) on page 37.
3. Updated “[Test Access Port – TAP](#)” on page 219 regarding the JTAGEN fuse.
4. Updated description for Bit 7 – JTD: JTAG Interface Disable on [page 228](#).
5. Added a note regarding JTAGEN fuse to [Table 104](#) on page 257.
6. Updated Absolute Maximum Ratings* , DC Characteristics and ADC Characteristics in “[Electrical Characteristics](#)” on page 287.
7. Added a proposal for solving problems regarding the JTAG instruction IDCODE in “[Errata](#)” on page 336.

Changes from Rev. 2503C-10/02 to Rev. 2503D-02/03

1. Added EEAR9 in EEARH in “[Register Summary](#)” on page 327.
2. Added Chip Erase as a first step in “[Programming the Flash](#)” on page 284 and “[Programming the EEPROM](#)” on page 285.
3. Removed reference to “Multi-purpose Oscillator” application note and “32 kHz Crystal Oscillator” application note, which do not exist.
4. Added information about PWM symmetry for Timer0 and Timer2.
5. Added note in “[Filling the Temporary Buffer \(Page Loading\)](#)” on page 251 about writing to the EEPROM during an SPM Page Load.
6. Added “Power Consumption” data in “[Features](#)” on page 1.
7. Added section “[EEPROM Write During Power-down Sleep Mode](#)” on page 22.
8. Added note about Differential Mode with Auto Triggering in “[Prescaling and Conversion Timing](#)” on page 204.
9. Updated [Table 89](#) on page 232.
10. Added updated “[Packaging Information](#)” on page 333.

Changes from Rev. 2503B-10/02 to Rev. 2503C-10/02

1. Updated the “[DC Characteristics](#)” on page 287.

Changes from Rev. 2503A-03/02 to Rev. 2503B-10/02

1. Changed the endurance on the Flash to 10,000 Write/Erase Cycles.
2. Bit nr.4 – ADHSM – in SFIOR Register removed.
3. Added the section “[Default Clock Source](#)” on page 25.
4. When using External Clock there are some limitations regards to change of frequency. This is described in “[External Clock](#)” on page 31 and [Table 117](#) on page 289.

5. Added a sub section regarding OCD-system and power consumption in the section [“Minimizing Power Consumption” on page 34](#).
6. Corrected typo (WGM-bit setting) for:
 - [“Fast PWM Mode” on page 75](#) (Timer/Counter0)
 - [“Phase Correct PWM Mode” on page 76](#) (Timer/Counter0)
 - [“Fast PWM Mode” on page 120](#) (Timer/Counter2)
 - [“Phase Correct PWM Mode” on page 121](#) (Timer/Counter2)
7. Corrected [Table 67 on page 164](#) (USART).
8. Updated V_{IL} , I_{IL} , and I_{IH} parameter in [“DC Characteristics” on page 287](#).
9. Updated Description of OSCCAL Calibration Byte.
 In the datasheet, it was not explained how to take advantage of the calibration bytes for 2, 4, and 8 MHz Oscillator selections. This is now added in the following sections:
 Improved description of [“Oscillator Calibration Register – OSCCAL” on page 30](#) and [“Calibration Byte” on page 258](#).
10. Corrected typo in [Table 42](#).
11. Corrected description in [Table 45](#) and [Table 46](#).
12. Updated [Table 118](#), [Table 120](#), and [Table 121](#).
13. Added [“Errata” on page 336](#).

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